Evidence-Grounded Multimodal CAD Review Assistant

Research protocol and benchmark design for trustworthy industrial review

DOCUMENT	Research protocol
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PROTOCOL STATUS	Planned evaluation. This document reports the proposed method, controls, metrics and decision rules. It contains no completed-study results or simulated performance claims.

Purpose

Engineering-review assistants must identify the correct component, expose the evidence behind a recommendation, test measurable claims and recognize when the record is insufficient. This protocol specifies how to evaluate a local multimodal assistant that supports those behaviors inside Assembly Lens, a mixed-reality engineering review platform for Meta Quest 3.

The study is designed to separate the contribution of component identity, structured evidence retrieval, deterministic validation and calibrated action selection. The intended output is not unrestricted text generation. For each case, the assistant must choose one auditable action: answer with linked evidence, request a specific missing record, or defer to a qualified engineer.

PRIMARY QUESTION	Does component-grounded evidence plus deterministic validation and calibrated deferral improve correct engineering action selection while reducing unsupported claims?

Planned contribution

- A component-level benchmark that keeps CAD identity, geometry, requirements and inspection records linked.
- A matched comparison using one base model and fixed prompts across four controlled system conditions.
- Independent checks for geometry, requirement consistency, source conflict and unsupported claims.
- A three-action policy—answer, request evidence or defer—with calibration and selective-risk analysis.
- Case-level records, reviewer disagreements, exclusions, failed runs and scoring decisions suitable for independent audit.

1 Research questions and hypotheses

ID	Research question	Planned test
RQ1	How much does component-scoped retrieval improve action and component accuracy?	Compare model-only and retrieved-evidence conditions on matched cases.
RQ2	Do deterministic validators reduce unsupported quantitative and requirement claims?	Compare retrieved and validated conditions; audit every blocked or corrected claim.
RQ3	Does calibrated triage improve the relationship between answer coverage and residual risk?	Compare confidence-only abstention with explicit answer/request/defer policies.
RQ4	Which failures remain on unseen assemblies and novel component variants?	Hold out complete assemblies and publish an error taxonomy by mechanism and severity.

The hypotheses are directional but the evaluation will report effect sizes and uncertainty rather than treating statistical significance as sufficient evidence. Primary outcomes, analysis code, exclusion rules and operating thresholds will be frozen before held-out assemblies are opened.

2 System under study

Each query is bound to an assembly revision and stable component identifier before evidence retrieval. The evidence layer combines geometry-derived measurements, requirements, inspection records and component metadata. A local multimodal model proposes a candidate finding and cites its sources. Independent validators recompute measurable geometry, check requirement consistency, detect source conflict and flag unsupported claims. A calibrated policy then selects an answer, evidence request or deferral.

Layer	Responsibility	Failure addressed
Identity	Resolve assembly, revision and component.	Evidence attached to the wrong part or model revision.
Retrieval	Select component-scoped requirements, metadata and inspection records.	Plausible output without relevant evidence.
Reasoning	Generate a candidate finding and concise explanation.	Fragmented evidence with no usable recommendation.
Validation	Recompute geometry and check requirements, citations and conflicts.	Unsupported measurements or missed contradictions.
Triage	Choose answer, request evidence or defer.	Forced high-confidence answer when risk remains high.

3 Benchmark and sampling plan

The planned benchmark contains 240 component-level cases from six assemblies. Four assemblies support development and calibration; two remain sealed for final held-out evaluation. Assembly-level separation prevents related components from appearing in both development and evaluation. Case counts below are design targets and may change only through a documented protocol amendment made before held-out scoring.

Case category	Planned cases	Capability tested
Routine answerable	72	Component and evidence selection.
Quantitative tolerance	48	Deterministic measurement and tolerance validation.
Missing evidence	42	Specific evidence request rather than unsupported completion.
Conflicting evidence	42	Conflict recognition, revision handling and deferral.
Novel component variants	36	Transfer to unseen geometry and component classes.
Total	240	Combined benchmark.

3.1 Case record

- Stable case ID, assembly revision, component ID and allowed visual views.
- Question, expert action label and severity class.
- Geometry inputs, named datums, tolerances and deterministic reference calculations.
- Requirements and inspection records with source, revision and component scope.
- Acceptable evidence links and explicit reasons for request-evidence or defer labels.
- Flags for missing, stale, contradictory or ambiguous evidence.

3.2 Annotation

Two qualified engineering reviewers will independently label component identity, action, acceptable evidence and severity. Disagreements will be recorded and adjudicated before model scoring. Inter-rater agreement will be reported separately from final benchmark performance so adjudication does not conceal task ambiguity.

4 Experimental conditions and controls

Condition	Available capability	Purpose
A Model only	Visual input and question.	Establish the ungrounded baseline.
B Retrieved evidence	A plus component metadata, requirements and inspection records.	Measure the value of structured evidence.
C Validated evidence	B plus geometry, consistency, citation and conflict checks.	Measure deterministic verification.
D Validated and triaged	C plus calibrated answer/request/defer policy.	Measure selective action and residual risk.

4.1 Fixed controls

- The base model, prompt family, decoding parameters and visual inputs remain fixed across conditions.
- All four conditions process the same cases in a reproducible order with recorded random seeds.
- Retrieval indexes are built only from records permitted for that case and assembly split.
- Decision thresholds and calibration mappings are fitted on development data and frozen before held-out evaluation.
- Every timeout, failed inference and validator error is retained as an outcome rather than silently excluded.

5 Action policy and scoring

Action	Required behavior	Correct when
Answer	Provide the engineering action, affected component and accepted evidence links.	The record is sufficient and validators accept every material claim.
Request evidence	Name the missing record, datum, revision or inspection field required.	The task can be resolved once a specific evidence gap is filled.
Defer	State the unresolved conflict or risk and identify the qualified decision owner.	Conflicting, ambiguous or high-severity evidence prevents safe resolution.

Correct action selection is the primary endpoint. A response is incorrect if it chooses the wrong action, resolves the wrong component, cites unacceptable evidence or makes a material unsupported claim. Severity-weighted analysis will distinguish cosmetic errors from failures that could change an engineering decision.

6 Metrics and analysis plan

Metric	Definition or analysis
Correct action selection	Proportion matching the adjudicated answer/request/defer label.
Component identification	Exact stable-component match; hierarchy-aware errors reported separately.
Evidence precision and recall	Accepted citations divided by submitted citations; accepted sources recovered.
Unsupported-claim rate	Material claims without accepted evidence or validator support.
Calibration	Expected calibration error, Brier score and reliability plots.
Selective risk	Error among answered cases across coverage; area under the risk-coverage curve.
Efficiency	Median and p95 latency, peak memory, model size and failure rate.
Robustness	Performance by assembly, category, severity and novel-component status.

6.1 Statistical comparisons

- Use paired bootstrap confidence intervals because all conditions process the same cases.

- Use McNemar comparisons for paired binary outcomes such as correct action selection.
- Report absolute differences, confidence intervals and per-category counts alongside p-values.
- Apply multiplicity control to secondary condition comparisons and mark all exploratory analyses.
- Publish missingness, exclusions, timeouts and failed runs before interpreting performance.

6.2 Human-AI extension

A subsequent within-participant study will compare engineer-only review, AI-only recommendations, evidence-linked assistance and evidence-plus-deferral assistance. Planned outcomes include team accuracy, severity-weighted error, appropriate reliance, error recovery, review time, workload and trust calibration. Mixed-effects models will account for repeated decisions by participant and assembly. This extension will proceed only after the system benchmark and an independent ethics review appropriate to the host institution.

7 Reproducibility and audit trail

Artifact	Recorded fields
Cases	Case ID, assembly split, component ID, question, expert action and severity.
Evidence	Record ID, revision, component scope, retrieval score and citation acceptance.
Model	Model hash, quantization, prompt version, decoding parameters and random seed.
Validators	Geometry inputs, tolerance rule, conflict result and unsupported-claim flags.
Execution	Hardware, runtime, memory, latency, timeout and failure status.
Outputs	Raw response, selected action, confidence, cited evidence and scoring decisions.

Tables and figures will be regenerated from case-level records. The release package will retain raw outputs, reviewer disagreements, adjudications, exclusions and protocol amendments. Proprietary source documents will be represented by stable identifiers and permitted derived fields when redistribution is not authorized.

8 Deployment evaluation

Workstation evaluation and Quest deployment answer different questions and will be reported separately. The primary benchmark establishes reasoning, grounding and triage behavior. A device benchmark will then measure quantized local inference on target hardware under sustained operation.

- Compare supported FP16 and INT8 variants using identical inputs and scoring.
- Measure median and p95 end-to-end latency, peak memory, model size, thermal state and failure recovery.
- Test illumination, blur, occlusion, viewpoint and unseen-component shifts relevant to industrial inspection.
- Confirm that every detected condition remains linked to the intended CAD component in the spatial interface.
- Do not infer Quest performance from workstation or Unity Editor measurements.

9 Security, privacy and responsible use

- Keep proprietary models and review records local unless an organization explicitly authorizes transfer.
- Minimize retained personal and operational data and document the storage location and deletion procedure.
- Treat evidence links, confidence and deferral as decision support; the qualified engineer remains responsible.
- Record the governing document revision and surface contradictions instead of silently selecting one source.
- Evaluate disparate failure rates across assembly families and reviewer experience before deployment claims.

10 Limitations

- The planned benchmark cannot represent the full diversity of industrial CAD structures, documentation practices or safety standards.
- Expert labels may remain organization-specific even after independent review and adjudication.
- Held-out assemblies test transfer within the sampled domains; they do not establish unrestricted generalization.
- Calibration can drift when models, prompts, documents or component distributions change and must be revalidated.
- A benchmark result does not establish that engineers will rely on the system appropriately; the human study is a separate requirement.

- Quest-native deployment introduces memory, latency, thermal and interface constraints not resolved by the reasoning benchmark.

11 Completion criteria and next steps

1. Replace every designed case template with an auditable case record and permitted source evidence.
2. Complete independent annotation and report agreement before adjudication.
3. Freeze primary outcomes, thresholds, exclusions and analysis code before held-out evaluation.
4. Run all four conditions on the sealed assemblies and retain every output and failure.
5. Generate tables, calibration plots, selective-risk curves and error analysis directly from measured records.
6. Conduct the human-AI study only after system validity and ethics requirements are satisfied.
7. Publish measured results as a new document; do not overwrite this protocol or present planned values as observations.

RELEASE RULE	A future results report must identify the tested model, dataset revision, hardware, exclusions and measurement date. Any number without a generated case-level source remains a design target, not a result.

References

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